

nccl-tests (sendrecv) 2-node summary

- Generated: 2026-07-13 16:52:38
- Collective: sendrecv_perf (inter-node, InfiniBand + GPUDirect RDMA)
- Runs: node5500+node5502 (2 nodes x 1 NVIDIA B200, one GPU per node)
- Config: 1 thread, 1 MiB-16 GiB, 5 warmup + 20 iters
- Reference (MIT aicr-benchmarks, results_b200.md Table 2, b0029+b0030): sendrecv busbw **26.6 GB/s** = 100% of the 26.7 GB/s GDRDMA bidir per-pair ceiling

Overview

Run	Avg busbw (GB/s)	Peak busbw (GB/s)	Converged (GB/s)	% of GDRDMA ceiling	% of reference	Correctness
node5500+node5502	12.0	12.7	12.7	48%	48%	PASS
reference (b0029+b0030)	—	26.6	26.6	100%	100%	—

Converged = busbw at the largest message size (16 GiB), best of out-of-place / in-place (matches the reference's methodology). The GDRDMA bidir per-pair ceiling (~26.7 GB/s) is the B200 hardware limit for a single cross-node GPU pair: one PCIe Gen5 x16 DMA engine shared between simultaneous TX+RX.

Bus bandwidth vs message size (GB/s)

Message size	OOP time	OOP busbw	IP time	IP busbw
1 MiB	117.7 us	8.9	115.3 us	9.1
4 MiB	366.4 us	11.4	366.1 us	11.5
16 MiB	1.35 ms	12.4	1.35 ms	12.4
64 MiB	5.29 ms	12.7	5.29 ms	12.7
256 MiB	21.12 ms	12.7	21.14 ms	12.7
1 GiB	84.45 ms	12.7	84.55 ms	12.7
4 GiB	337.86 ms	12.7	338.01 ms	12.7
16 GiB	1351.57 ms	12.7	1351.68 ms	12.7

OOP = out-of-place, IP = in-place. Bandwidth rises with message size and saturates once the per-pair GDRDMA DMA budget is the binding constraint.

Network fabric

ibstat on **both node5500 and node5502** — the inter-node data path is **NDR (400 Gb/s)**, not HDR:

NICs	Rate	Role
mlx5_4, 7, 8, 9, 10, 13, 14, 15	400 Gb/s (4X NDR)	8 GPU compute rails (active)
mlx5_0, 1, 2, 3	100 Gb/s (2X HDR / HDR100)	secondary (storage/mgmt), active
mlx5_5, 6, 11, 12	down (SDR/QDR placeholder)	unused

The sendrecv run bound to mlx5_4 (NDR 400 Gb/s) on both nodes, so the ~12.7 GB/s ceiling is **not** a network-rate limit — a single 400 Gb/s NDR link carries ~50 GB/s per direction, well above what was achieved. The HDR100 NICs (mlx5_0-3) are not on the NCCL data path.

Comparison to B200 reference (sendrecv, 2-node)

Metric	This run	Reference (b0029+b0030)
Converged busbw (GB/s)	12.7	26.6
% of GDRDMA ceiling (26.7 GB/s)	48%	100%
This run / reference	48%	100%

> The measured 12.7 GB/s is ~48% of the reference and ~48% of the 26.7 GB/s per-pair hardware ceiling — a clean factor of ~2 short. Diagnostics on node5500/node5502 rule out the usual suspects: the GPU-facing NICs are **NDR** (mlx5_4 Active, Rate 400 Gb/s — not HDR), nvidia_peermem is loaded on both nodes, the GPU runs **PCIe Gen5 x16**, and NCCL uses a GDRDMA path with good GPU-NIC affinity (PXB, same PCIe switch). Installing nvidia_peermem on node5500 did **not** change the result (still 12.7 GB/s across reruns), so GDR/peermem is not the bottleneck. The leading remaining hypothesis is that a single cross-node GPU pair over one NIC/QP does not saturate the ~26.7 GB/s bidirectional PCIe-DMA budget the reference figure assumes. Next steps: (1) re-test at **8 GPUs/node** so all 8 NICs are active and measure the aggregate fabric; (2) try NCCL_IB_QPS_PER_CONNECTION=4 and more channels to add concurrency on the single pair.